

Minimal Edits, Maximum Privacy: Bayesian-Guided Generative Agents for Silent Geolocation Shielding

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Abstract—The rapid evolution of Vision-Language Models (VLMs) and geolocation techniques has precipitated widespread geoprivacy concerns amidst the explosion of social media imagery. While countermeasures exist, current protection techniques typically rely on naive obfuscation or adversarial perturbations, and these methods often compromise visual aesthetics and consistency while offering limited robustness against localization. To address this gap, we introduce a fully automated pipeline for generating privacy-preserving images via minimal, risk-aware modifications. Our approach leverages feature extraction methods combined with a Bayesian network to identify vulnerable visual factors, utilizing a VLM-driven Edit-Agent Loop to guide generative editing. This ensures modifications are semantically targeted to neutralize risks while maintaining image integrity. Experiments demonstrate that our framework achieves robust protection with edits that remain virtually imperceptible. Consequently, our work offers a fresh perspective on safeguarding image geoprivacy in the era of Large Language Models.

Index Terms—Geoprivacy, Privacy-Preserving Image Editing, Risk Assessment, Dataset Generation, Generative AI

I. INTRODUCTION

The proliferation of social media platforms and high-resolution smartphone cameras has led to a substantial increase in social media imagery, which serves as the primary medium for communication and self-expression [1]. While this rich data stream facilitates social interaction, it simultaneously exposes users to severe, often underestimated passive privacy risks. Recent studies have investigated these risks across several dimensions, such as facial privacy [2], bystander privacy [3], and private attribute inference [4]. Beyond these identifiable subjects, the risk of implicit location leakage through social media images demands urgent attention.

Images shared on social media inherently carry severe geoprivacy risks, potentially resulting in economic damage or physical harm [5]. However, many users remain largely unconcerned or unaware of the hidden visual cues that compromise their location [6]. This lack of awareness is continuously exploited by increasingly sophisticated geolocation techniques, which means, to infer an image's location through implicit visual cues. It is powered by Open Source Intelligence (OSINT) [7], Geoguessr communities [8], and further amplified by advanced AI models [9] and Visual Large Language Models

(VLMs) [10], [11]. Through these increasingly powerful tools, malicious entities can now infer precise locations from seemingly innocuous visual details. Consequently, the convergence of user desensitization and powerful automated surveillance has catalyzed a significant increase in unintentional location data leaks [12].

To mitigate the risk of implicit location leakage, researchers have explored various image modification strategies. Hasan et al. [13] evaluated the trade-off between privacy preservation and user experience using rudimentary methods like cropping and masking, yet concluded that these approaches yield unsatisfactory results for both objectives. With the rise of deep learning, subsequent studies [14], [15] utilized adversarial perturbations to deceive AI geolocation models. However, these techniques often introduce visible artifacts and lack robustness against adversarial purification [16]. More recently, driven by advancements in Visual Large Language Models (VLMs), Ma et al. [17] proposed using Large Language Models to identify and blur risky visual elements, yet their approach still results in aesthetic degradation. In conclusion, existing image geolocation protection methods often significantly compromise the semantic understanding and interpretability of images, while failing to effectively safeguard geoprivacy.

To address these challenges, we introduce a novel, fully automated pipeline for **unobtrusive geolocation shielding**. Unlike prior works that rely on obfuscation, our approach treats privacy protection as a **decoupling process** between visual cues and coordinates via a **Bayesian-Guided Generative Agent Loop**. First, a pre-trained Bayesian Network analyzes extracted multi-dimensional features to diagnose their contribution to localization probability, generating a quantitative Risk Report. Subsequently, an iterative agent loop neutralizes these risks: (1) A VLM agent maps risk factors to segmented scene elements to formulate surgical editing instructions; (2) A generative model executes these edits; (3) An adversarial "Red Team" model validates the output. If geolocation succeeds ($<100\text{km}$), its reasoning trace is injected back to refine the instructions in (1). Extensive experiments demonstrate that our method effectively safeguards geoprivacy against advanced AI models while maintaining superior visual fidelity. Consequently, this work offers a fresh perspective on safeguarding image geoprivacy in the era of Large Language Models, and establishes a new paradigm for agentic privacy

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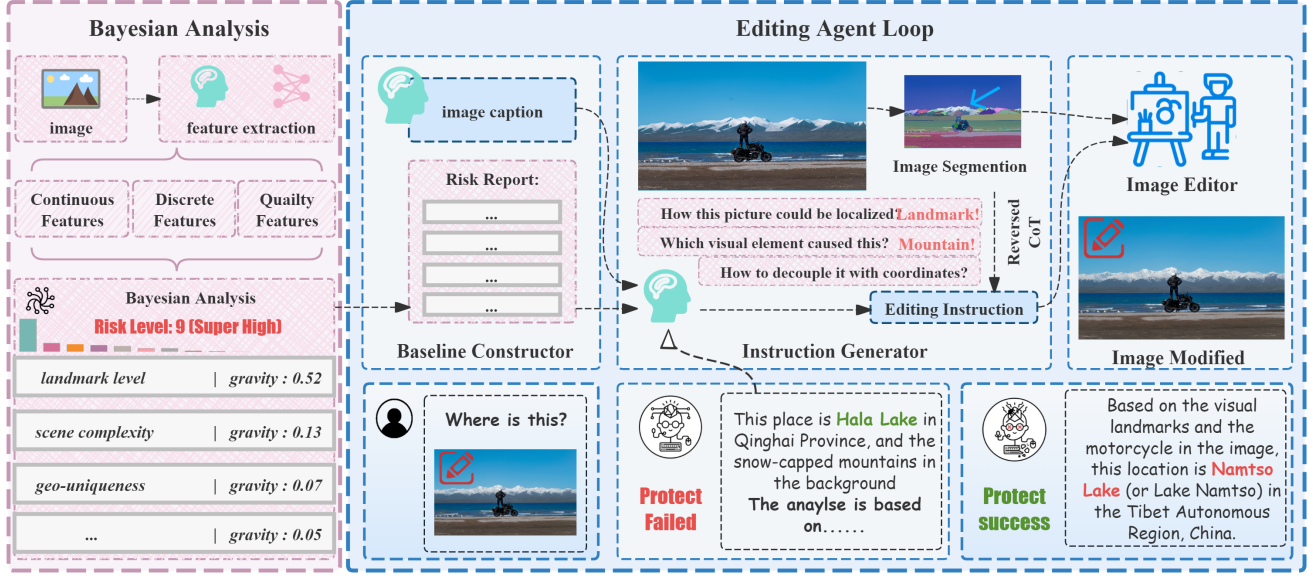


Fig. 1. **Overview of the Bayesian-Guided Generative Pipeline.** The workflow proceeds as follows: (1) **Bayesian Analysis.** Multi-granular features are analyzed by a Bayesian Network to generate a quantitative Risk Report. (2) **Editing Agent Loop.** This loop consists of four phases: **Semantic Mapping:** SAM decomposes the scene into object-level masks for spatial precision. **Agent Reasoning:** A VLM Agent correlates risk factors with masks to formulate “surgical” editing instructions. **Generative Execution:** The image is modified unobtrusively within target masks. **Adversarial Feedback:** A high-performance VLM acts as a “Red Team” attacker to geolocate the output. If successful (Error<100km), its reasoning trace is captured and injected back into the generator as a Reversed Chain-of-Thought (Reversed CoT) to guide the system to iteratively refine the editing strategy.

preservation.

Our main contributions are summarized as follows:

- We propose a novel automated framework that integrates quantitative risk assessment with generative AI agents. This “Edit-Agent Loop” enables the system to iteratively reason about and neutralize privacy threats.
- We introduce a greedy-like optimization strategy powered by the Agent Loop to resolve the geolocation privacy dilemma, effectively balancing robust privacy protection with the preservation of image aesthetics.
- Extensive experiments demonstrate that our pipeline significantly reduces the geolocation success rate of VLM and geolocation AI models. Furthermore, ablation studies confirm that the introduction of the Red Team adversarial mechanism and Bayesian risk analysis contributes significantly to enhancing protection success rates while maintaining image consistency.

II. METHODOLOGY

A. Problem Formulation

We address the problem of protecting visual geoprivacy against advanced AI-powered geolocation models and human experts. Let $f_{loc} : \mathcal{I} \rightarrow \mathcal{Y}$ be any geolocation function (e.g., GeoCLIP, VLMs) that maps an image $I \in \mathcal{I}$ to a geographic coordinate or region label $y \in \mathcal{Y}$. Our goal is to generate a protected image I' such that the geolocation model is misled, while preserving the original visual aesthetics and

semantic integrity. We formulate the objective as maximizing the geolocation divergence:

$$\max_{I'} \mathcal{D}(f_{loc}(I), f_{loc}(I')) \quad \text{s.t.} \quad \text{Sim}(I, I') > \tau \quad (1)$$

where $\mathcal{D}$ represents the divergence metric (e.g., distance in embedding space or Cross-Entropy loss against the ground truth), and Sim ensures perceptual fidelity.

To achieve this “unobtrusive shielding,” we propose a pipeline shown in Fig. 1.

B. Stage 1: Bayesian-Guided Feature Attribution

To guide the editing process, we first interpret the implicit geographic cues within the image. To achieve this, we construct a Bayesian Network (BN) to model the probabilistic dependency between visual features and geolocation vulnerability.

1) *Feature Extraction and Risk Model:* Cheng *et al.* [18] constructed a comprehensive Bayesian Network (BN) to quantitatively assess the implicit geoprivacy risk of User-Generated Content. Their work introduced a feature set of 11 metrics across four categories, modeling the causal pathways from visual features to specific geolocation vulnerabilities. Building upon this foundational framework, we select 9 image-specific features (excluding non-visual metadata) to evaluate our inputs. These include continuous geographical features (e.g., GeoCLIP [19] similarity mean and variance), discrete semantic features (e.g., text value, landmark level, and scene complexity), and image quality attributes (e.g., weather, time of day, view angle, focal length, and clarity).



Fig. 2. **Case Study of the Bayesian-Guided Edit Agent Loop.** Representative samples from the IM2GPS3k dataset are processed by our pipeline to generate privacy-preserved counterparts. We employ *Gemini-2.5-Pro* to attempt geolocation on these outputs, demonstrating the proposed pipeline’s effectiveness in neutralizing implicit location cues.

The core function of our BN module is to perform **Risk Attribution**. Using the pre-trained BN structure from [18] fitted on the Chaofun42K dataset [20], we employ a sensitivity-based forward inference mechanism to quantify the marginal contribution of each visual feature to the overall geolocation vulnerability:

Let $\mathbf{E} = \{e_1, e_2, \dots, e_N\}$ denote the set of observed evidences for a given image I . The baseline geolocation risk R_{total} is the conditional probability of the target node Y being in the “Solved” state given the full evidence:

$$R_{total} = P(Y = \text{High} \mid \mathbf{E}) \quad (2)$$

To identify the **Dominant Risk Factors**, we perform an iterative ablation analysis. For each feature i , we define a subset of evidence $\mathbf{E}_{-i} = \mathbf{E} \setminus \{e_i\}$, where the evidence for feature i is removed (effectively reverting it to its prior distribution). The importance score $\mathcal{I}(e_i)$ is calculated as the absolute deviation caused by this removal:

$$\mathcal{I}(e_i) = |P(Y = \text{High} \mid \mathbf{E}) - P(Y = \text{High} \mid \mathbf{E}_{-i})| \quad (3)$$

By ranking all features based on $\mathcal{I}(e_i)$, we generate a structured **Risk Attribution Report** $\mathcal{R}_{attr}$. This report provides the Agent with an ordered list of “targets” that yield the highest return on privacy investment.

C. Stage 2: Holistic Semantic Mapping

To translate the abstract $\mathcal{R}_{attr}$ into actionable spatial regions, we employ the **Segment Anything Model (SAM)** [21]. SAM decomposes the image I into a set of disjoint semantic masks $M = \{m_1, m_2, \dots, m_n\}$. This provides a pixel-level index map, allowing the agent to associate a conceptual risk (e.g., “high-value text”) with a specific spatial region (e.g., “Mask #4”).

D. Stage 3: The Adversarial Edit-Agent Loop

This stage constitutes the core of our pipeline, integrating reasoning, execution, and verification into a unified adversarial

framework. We deploy a VLM Agent acting as a Privacy Specialist to orchestrate the protection process. This process operates iteratively, treating privacy protection as a game between a **Defender** (the editing agent) and a **Red Team Attacker**.

1) *Agent Reasoning and Strategy Formulation:* The Defender Agent operates on a strict Chain-of-Thought (CoT) protocol defined by our system prompt (is shown in supplementary material). It receives the tuple $(I, \mathcal{R}_{attr}, M)$, where $\mathcal{R}_{attr}$ is the Bayesian Risk Report and M is the semantic mask set. The agent executes a three-step process to generate the editing plan:

- 1) **Risk Source Analysis:** The agent analyzes $\mathcal{R}_{attr}$ to filter factors with high risk scores. It prioritizes “High-Uniqueness” cues (e.g., specific landmarks) over generic ones.
- 2) **Spatial Grounding:** The agent correlates these abstract risks with the Semantic Map M .
- 3) **Strategy Selection:** The agent selects from a defined knowledge base of **Editing Actions** to decouple the location link while preserving aesthetics. The action space includes:
 - **“Appearance Alteration”:** Changing textures or materials.
 - **“Semantic Replacement”:** Swapping a unique object with a generic equivalent.
 - **“Visual Obfuscation”:** Adding natural environmental effects.

The output is a target mask ID m_{target} and a natural language instruction T_{edit} (e.g., “Blur the text on the sign naturally (Mask #12) to remove address information”).

2) *Generative Execution and Red Team Verification:* The instruction is executed by a generative inpainting model G (*gemini-2.5-flash-image*), producing a candidate image $I^{(t)} = G(I^{(t-1)}, m_{target}, T_{edit})$. To ensure robustness, we introduce an **Adversarial Feedback Loop**:

TABLE I

QUANTITATIVE EVALUATION OF GEOLOCATION SHIELDING EFFICACY. WE EVALUATE THE PERFORMANCE OF GEOLOCATION MODELS AND VLMS ON THE IM2GPS3k AND YFCC4k BENCHMARKS. THE METRICS REPORT THE PERCENTAGE OF IMAGES LOCALIZED WITHIN A SPECIFIC DISTANCE THRESHOLD (LOWER IS BETTER FOR PRIVACY) AND THE MEAN ERROR (HIGHER IS BETTER FOR PRIVACY).

Benchmark	Evaluator Model	Setting	Avg Err. (km) ↑	Localization Accuracy (% within Distance)				
				1 km ↓	25 km ↓	200 km ↓	750 km ↓	2500 km ↓
IM2GPS3k (filtered)	GeoCLIP (ViT-L/14)	Unprotected Ours	1167.84 2035.26	19.15 9.57	47.87 25.00	62.23 40.43	77.66 62.23	88.83 79.26
	Qwen3-VL-30B	Unprotected Ours	744.36 1715.65	14.52 7.53	47.85 29.03	68.28 44.62	86.02 68.82	94.62 82.80
	Gemini-2.5 Pro	Unprotected Ours	360.66 1419.37	30.98 11.96	73.37 32.98	82.61 48.91	91.30 73.91	96.74 86.96
	GPT-5	Unprotected Ours	272.27 1787.31	32.42 13.87	72.53 39.88	82.42 51.45	89.56 68.79	97.80 80.92
YFCC4k (filtered)	GeoCLIP (ViT-L/14)	Unprotected Ours	1514.37 2295.26	15.83 9.17	35.00 21.67	47.50 32.50	74.17 57.50	85.00 75.83
	Qwen3-VL-30B	Unprotected Ours	1945.24 2206.44	7.27 5.45	23.64 14.55	46.36 33.64	67.27 55.45	77.27 71.82
	Gemini-2.5 Pro	Unprotected Ours	702.87 1778.18	35.24 13.59	60.95 26.21	72.38 44.66	86.67 69.90	92.38 82.52
	GPT-5	Unprotected Ours	1778.71 1941.88	28.71 19.75	58.42 35.80	68.32 48.15	77.23 70.37	79.21 81.48

The Red Team Attack: We deploy a high-performance VLM (*Gemini-2.5 Pro*) as a proxy attacker $\mathcal{A}$. It attempts to geolocate $I^{(t)}$ and provides a reasoning trace $\mathcal{T}_{reason}$.

Reversed Chain-of-Thought (Reversed CoT): If the defense fails (Dist $\geq$ 100km), the Defender Agent ingests the attacker’s $\mathcal{T}_{reason}$ to perform Logic Inversion. Instead of using the reasoning to find the location, the agent uses it to hide it. The instruction generator refines $\mathcal{T}_{edit}$ to specifically target the “leak” identified by the attacker.

This loop continues until the Red Team fails to geolocate the image or a maximum iteration count is reached.

III. EXPERIMENTS

A. Experiment Setup

a) Datasets: We utilize two datasets: **IM2GPS3k** and **YFCC4k**. To evaluate the robustness of our protection on highly vulnerable content, we employed the Bayesian Network described in Stage 1 to pre-filter the datasets, selecting only images with a vulnerability score exceeding 0.7. This selective process explains why the baseline geolocation accuracies reported in our table surpass typical state-of-the-art benchmarks. Our primary objective is to rigorously test the robustness of our defense method rather than to benchmark the general geolocation capabilities of VLMS. The ablation study relies on the **Im2gps3k-filtered** dataset.

b) Evaluation Metrics: To compare the effectiveness of our method with naive baselines, we employ the same image modification model (Gemini-2.5-Flash-Image) across three distinct modification strategies. **LPIPS** measures perceptual distance, **PSNR** measures pixel fidelity, and **CLIP-I** measures

TABLE II

COMPARISON OF EFFECTIVENESS. QUANTITATIVE COMPARISON OF VISUAL FIDELITY AND PRIVACY PROTECTION BETWEEN OUR METHOD AND NAIVE BASELINES ON THE FILTERED IM2GPS3k DATASET.

Method	LPIPS ↓	PSNR (dB) ↑	Loc Acc @25km ↓	FID ↓
<i>Original Image</i>	0.000	∞	1.000	2.15
Naive VLM Blurring	0.248	30.33	62.50	89.36
Generative Replacement	0.311	29.96	52.73	79.17
Ours (Loop Edited)	0.288	30.36	32.98	74.90

semantic consistency. **FID** (Fréchet Inception Distance) assesses the realism and distribution distance from real images. In Table I, we report both **Localization Accuracy** (percentage of images localized within a specific distance threshold) and **Mean Error** (average distance between predicted and ground truth coordinates). Note that Accuracy reflects precise localization capabilities, while Mean Error captures the magnitude of deviation when localization fails.

B. Main Results: The Privacy-Utility Trade-off

We evaluated our pipeline’s ability to reconcile the conflicting objectives of *Maximum Privacy* and *Minimal Edits*.

1) Maximum Privacy: Neutralizing Fine-Grained Geolocation: Our pipeline delivers a decisive blow to the precision of advanced geolocation methods. On the filtered IM2GPS3k benchmark, the **GPT-5** model originally achieved **32.42%** accuracy within 1 km. After processing, this dropped to **13.87%**. Similarly, the specialized **GeoCLIP** model saw its 1 km accuracy decrease from **19.15%** to **9.57%**. Critically, the **Mean Error** increased significantly (e.g., GPT-5 from 272.27



Fig. 3. **Visualization of the Edit-Agent Loop and Attention Shift.** We illustrate the progressive modification process where the image is iteratively altered until geolocation fails. The second row visualizes the attention map of the *Qwen3-VL-30B* model during geolocation, computed using the method from [22].

TABLE III
ABLATION STUDY ON KEY COMPONENTS. WE EVALUATE THE CONTRIBUTION OF THE BAYESIAN RISK MODULE AND THE SEMANTIC SEGMENTATION MODULE. *Geolocation Accuracy* REFERS TO THE SUCCESS RATE OF THE ATTACKER *Gemini-2.5-Pro* WITHIN 25KM.

Method	Geo. Acc. (25km) ↓	PSNR (dB) ↑	FID ↓
Original Image	73.37%	∞	2.15
w/o Bayesian (Generic VLM)	58.06%	30.59	84.70
w/o Segmentation (Global Edit)	52.00%	30.36	72.43
Full Pipeline (Ours)	32.98%	30.36	74.90

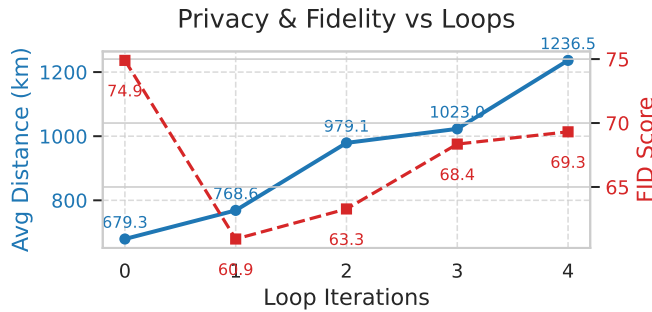


Fig. 4. **Efficacy of the Edit-Agent Loop with Reversed CoT.** The chart illustrates the trade-off between Defense Success Rate (Left Axis, Blue) and Visual Fidelity (Right Axis, Red, measured by FID) as the maximum loop iterations (T) increase. Ideally, higher Defense Rate and lower FID are preferred. The results show that the *Reversed CoT* mechanism significantly boosts protection in the early stages ($T = 1$ to 3) by addressing “hard” semantic cues, reaching an optimal balance at $T = 3$ before diminishing returns set in.

km to 1787.31 km), confirming that the semantic link between visual cues and coordinates was effectively disrupted.

We observe that our method demonstrates stronger protection against VLMs (85% mean error decrease on *GPT-5*) compared to models like *GeoCLIP* (43% mean error decrease), which are trained on massive image-coordinate pairs. This discrepancy likely arises because our approach prioritizes modifying subtle yet effective semantic elements, rather than broadly altering low-level statistical features that such discrim-

inative models rely on.

Notably, the accuracy at coarse scales (e.g., @ 2500 km) remains high ($\sim 40\%$). This a design feature: unlike naive black-box methods that obliterate random context, our agent preserves continuous features to maintain image naturalness, decoupling the specific location identity while retaining the scene’s general atmosphere.

2) *Minimal Edits: Silent and Unobtrusive Shielding*: Table II validates the effectiveness of our protection compared to *Naive Blurring* and *Generative Replacement* under the same model backbone on our filtered IM2GPS3k dataset.

- **Pixel-Level Fidelity**: Our method achieves the highest PSNR (30.36 dB), indicating superior retention of original pixel information compared to the baselines.
- **Perceptual Realism**: In terms of naturalness, our method achieves the lowest FID (74.90), significantly outperforming both *Naive Blurring* (89.36) and *Generative Replacement* (79.17).
- **Semantic Consistency**: While *Generative Replacement* achieves the highest CLIP-I (0.916) due to unconstrained generation, our method maintains a comparable score (0.894).

C. Ablation Study

1) *Impact of Key Components*: To validate the necessity of each module, we conducted an ablation study as shown in Table III.

- **Effect of Bayesian Risk Attribution**: Removing the Bayesian module (“w/o Bayesian”) forces the VLM to blindly guess which elements are risky. It fails to match the privacy protection of our Full Pipeline (Acc@25km: 58.06% vs 32.98%). More importantly, without the priority ranking from the risk report, the agent often edits “obvious” but low-weight features (e.g., cars) while missing subtle environmental cues, or over-edits the scene, resulting in the worst FID (84.70).
- **Effect of Semantic Segmentation**: Removing the SAM-based mask constraint (“w/o Segmentation”) turns the

operation into a global image-to-image translation. Although this maintains competitive visual fidelity (FID 72.43), it significantly compromises privacy protection (Acc@25km: 52.00% compared to 32.98% for Ours). This confirms that segmentation is crucial for "Unobtrusive" yet effective shielding, allowing the agent to focus edits on high-risk regions.

2) *Efficacy of the Edit-Agent Loop*: A core innovation of our pipeline is the adversarial feedback loop. To evaluate the effectiveness of this loop, we measured the defense success rate (percentage of images where geolocation fails) and visual quality (FID) across different maximum loop iterations (T), as shown in Fig. 4.

One-Shot vs. Iterative Refinement: At $T = 1$ (single-pass editing), the system achieves a moderate defense rate. However, stubborn features often remain—for example, a VLM attacker might reason: "The street sign is blurred, but the unique mountain ridge in the background confirms this is Cape Town." By enabling the loop ($T = 2, 3$), our instruction generator ingests this attack log. Utilizing Reversed CoT logic, it generates a new directive: "The attacker identified the mountain ridge as a leak. Apply volumetric fog to the mountain region." This refinement boosts the defense success rate to **85.11%** (Acc@1km: 14.89%) at $T = 3$.

Convergence: We observe diminishing returns after $T = 3$, while the FID score continues to rise slightly as more pixels are altered. This indicates that a maximum loop of 3 strikes the optimal balance between robust security and aesthetic preservation.

IV. CONCLUSION

In this paper, we address the critical challenge of protecting implicit geoprivacy against advanced geolocation methods and VLMs. We propose a novel **Silent Geolocation Shielding** pipeline that leverages a **Bayesian-Guided Generative Agent Loop** to decouple visual cues from location identities. By integrating quantitative risk assessment with an adversarial feedback mechanism featuring **Reversed Chain-of-Thought**, our system iteratively refines surgical edits to neutralize high-risk elements without compromising image aesthetics. Extensive experiments confirm that our approach effectively thwarts geolocation models while preserving superior visual fidelity. Furthermore, our risk attribution model relies on semantic causal relationships (e.g., landmark identities to locations) rather than low-level pixel distributions, demonstrating robust cross-dataset generalization across diverse real-world distributions. In the future, we plan to construct a large-scale privacy-preserving dataset using this pipeline to train lightweight, end-to-end models for edge deployment.

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